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\\Research

Investigators

Frederick

NAV_optionscan_P1 (2)

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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T21_Protocol_filename_Generic

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	BC

Contrast - Common

TR	11.0 ms
TE	4.80 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	8.000

Physio - Signal

1st Signal/Mode	None
TR	11.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
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Sequence - Part 1

Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	Generic
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL01_unchanged_ep_moco_nav_set_AB

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	BC

Contrast - Common

TR	11.0 ms
TE	4.80 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	8.000

Physio - Signal

1st Signal/Mode	None
TR	11.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
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Sequence - Part 1

Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	Generic
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Sequence - Assistant

SAR Assistant	Off
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\\Research\Investigators\Frederick\NAV_optionscan_P1 (2)\CTRL02_unchanged_ep_moco_nav_set_AB

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	BC

Contrast - Common

TR	11.0 ms
TE	4.80 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	8.000

Physio - Signal

1st Signal/Mode	None
TR	11.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
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Sequence - Part 1

Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	T2-SPACE
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T01_Readout_polarity_Negative

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Negative
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T02_Nav_location_After

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Negative
Nav. location	After
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T03_Apply_moco_to_neither

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	After
Apply moco to	neither

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\Investigators\Frederick\NAV_optionscan_P1 (2)\T04_Remeasure_50

TA: 6:09 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	None
Apply moco to	parent and nav

Sequence - Special

Remeasure	0 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	Off
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T05_Reacq_threshold_055

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	neither

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.55
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T06_Feedback_Delay_610

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	61 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\Research\Investigators\Frederick\NAV_optionscan_P1 (2)\T07_Moco_ref_image_NewSessRef

TA: 7:14 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
-----------------	------

Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	25 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	New Sess Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T08_K_space_streaming_File

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.55
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	File
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T09_ABCD_navigator_Off

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	61 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	Off
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T10_Apply_freq_to_neither

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	New Sess Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	neither
Averaging	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T11_Add_grad_time_01

TA: 1 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	3.00 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	3.00 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Sequence - Part 1

Sequence Name	
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav
Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Sess Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.10 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL03_unchanged_tfl_mgh_epinav_ABC

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	File
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL04_unchanged_tfl_mgh_epinav_ABC

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	Network
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL05_unchanged_tfl_mgh_epinav_ABC

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	Off
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL06_unchanged_tfl_mgh_epinav_ABC

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav

Sequence - Special

Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	neither
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\Investigators\Frederick\NAV_optionscan_P1 (2)\T12_Include_Nav__Off

TA: 5:34 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
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Sequence - Special

Include Nav.	Off
Apply moco to	parent and nav
Remeasure	0 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	Off
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T13_Apply_moco_to_neither

TA: 5:34 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	Off
Apply moco to	neither
Remeasure	0 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T14_Remeasure_190

TA: 6:38 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	neither
Remeasure	19 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T15_Reacq_threshold_055

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	nav only
Remeasure	18 TRs
Reacq. threshold	0.55
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T16_Feedback_Delay_810

TA: 6:38 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	19 TRs
Reacq. threshold	0.50
Feedback Delay	81 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\NAV_optionscan_P1 (2)\T17_Moco_ref_image_NewSessRef

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.55
Feedback Delay	80 ms
Moco ref. image	New Sess Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T18_K_space_streaming_File

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	81 ms
Moco ref. image	Use Temp Ref
K-space streaming	File
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T19_ABCD_navigator_Off

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	New Sess Ref
K-space streaming	None
ABCD navigator	Off
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\T20_Apply_freq_to_neither

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Sess Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	neither

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL07_unchanged_space_mgh_epinav_A

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	File
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL08_unchanged_space_mgh_epinav_A

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
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Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	Network
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\NAV_optionscan_P1 (2)\CTRL09_unchanged_space_mgh_epinav_A

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
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Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	Off
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators\\Frederick\\NAV_optionscan_P1 (2)\\CTRL10_unchanged_space_mgh_epinav_A

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249571 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
--------------	----

Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	neither

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s